TOSHKENT SHAHRIDAGI INHA UNIVERSITETI
INHA UNIVERSITY IN TASHKENT

–1 POINTS FOR EACH FIELD LEFT EMPTY OR FILLED WRONG

FIRST NAME:		LAST NAME:	
ID NUMBER:		GROUP/SECTION:	
STUDENT SIGNATURE:		ROOM NUMBER	

MIDTERM EXAMINATION

COURSE NAME:	DISCRETE MATHEMATICS	COURSE NUMBER:	MSC 2050
EXAMINATION DURATION:	1 HOUR 30 MINUTES		
ADDITIONAL MATERIALS ALLOWED TO BE USED:	CALCULATOR		
SPECIAL INSTRUCTIONS	(1) FOLLOW ANY INSTRUCTIONS GIVEN BY PROCTORS. COMPLIANCE IS REQUIRED! DO NOT ARGUE WITH A PROCTOR, YOU GET "F" FOR MISSBEHAVING + other possible complications. (2) ONCE YOU ARE ALLOWED TO OPEN THE EXAMINATION BOOKLET, YOU MUST PUT ONLY YOUR **SIGNATURE** ON EVERY ODD PAGE (do NOT write your name/ID number inside the examination paper)		

- Final answers must be written by **only blue or black, non-erasable pen**.
- Do NOT use highlighters or correction pen.
- IUT's Examination Rules and Regulations apply.

Do NOT open the examination paper until directed to do so!

1. (8 points) Consider a proposition $\neg(p \rightarrow \neg q)$.

   (a) (2 points) Construct the truth table of this proposition.

   (b) (1 point) Using the truth table or otherwise show that $\neg(p \rightarrow \neg q) = p \wedge q$.

   (c) (3 points) Using the notations
   s="The diagnostic message is stored in the buffer"
   r="The diagnostic message is re-transmitted"
   Translate the following propositions into logical expressions:
   A) " It is not the case that if the diagnostic message is stored in the buffer, then it is re-transmitted."
   B) "The diagnostic message is stored in the buffer and it is not re-transmitted."
   C) "The diagnostic message is not stored in the buffer only if it is re-transmitted."

   (d) (2 points) Determine whether these system specifications are consistent:
   It is not the case that if the diagnostic message is stored in the buffer, then it is re-transmitted. The diagnostic message is stored in the buffer and it is not re-transmitted. The diagnostic message is not stored in the buffer only if it is re-transmitted.
   NOTE: there are two equivalent system specifications.

2. (8 points)

(a) (2 points) Translate the following statement to an English sentence

$$\forall m \forall n \exists p((m < p) \wedge (p < n))$$

(b) (2 points) Show that the statement in (a) is false over the domain of real numbers.

(c) (2 points) Change the statement in (a) (restrict the domain) so that the new statement is true over the domain of real numbers.

(d) (2 points) Change the statement in (a) (restrict the domain) so that the new statement is true over the domain of integers.

3. (10 points) Consider the sets

$$A = \{-1, 2, 4, 6\}$$

$$B = \{n \mid n \text{ is integer and } -2 \leq n \leq 5\}.$$

(a) (1 point) Find $A \cup B$ and $B \cap A$.

(b) (1 point) Find $A - B$ and $B - A$.

(c) (1 point) Find $|A|$ and $|B|$.

(d) (1 point) Find $\mathcal{P}(A)$

(e) (1 point) Construct any function from A to B

(f) (1 point) Find domain and range of the function in (e)

(g) (1 point) Find codomain of the function in (e)

(h) (1 point) Is the function in (e) one-to-one? Justify.

(i) (1 point) Is the function in (e) onto? Justify.

(j) (1 point) Is the function in (e) bijection? Justify.

4. (6 points) (a) (4 points) Write an algorithm in pseudocode, which counts number of 1s in a given bit string of length n.

   For example, in the case when the bit string is 10010 your algorithm must return 2.

   (b) (2 points) Find the worst and average time complexities of your algorithm from (a).

5. (8 points) Consider the affine cypher on the English alphabet given by the encryption function $e(x) = (3x + 1) \,\mathbf{mod}\, 26$.

   (a) (2 points) Find the inverse of 3 modulo 26.

   (b) (2 points) Find the decryption function.

   (c) (3 point) Decrypt the encrypted message "ORGWZOT ZD ZLURDDZEIN". The alphabet below may be helpful.

A	B	C	D	E	F	G	H	I	J	K	L	M
$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$	$\updownarrow$
13	14	15	16	17	18	19	20	21	22	23	24	25

   (d) (1 points) Explain why the decryption is not possible in case when the encryption function is $e(x) = (13x + 2) \,\mathbf{mod}\, 26$.

Do NOT fill	P.1	P.2	P.3	P.4	P.5	Total
max points	8	8	10	6	8	40
your score						